Class 16 project + limit cycles

Activity

Teams 5 & 6, post photos of your work to the class Google Drive (see Canvas for link). Make a folder for today's class if one doesn't exist yet.

Projects

Your final project will be based on a published paper or book chapter. The goal of the project is to understand and replicate key results of the paper or chapter, and to try a small extension or different exploration.

Your team will have two project presentations. A progress presentation in the last week of class and a final presentation on Thursday May 7th from 2-5pm.

Goals of the progress presentation:

- Introduce your project topic to your classmates in a way that they can understand
- Connect your project to the mathematical content of the course
- Present your project goals
- Introduce new mathematical content in the project (from beyond the scope of our course)
- Share your progress
- Illustrate your ideas to make them more accessible

Goals of the final presentation:

- Provide a reminder of project question/goal
- Provide background your classmates will need to understand your work (incorporating feedback on your progress presentation)
- Present the approach/model that you were replicating.
- Share the results of your replication work and your conclusions
- Provide recommendations for next steps (where you would take this project if you had more time)

1. Project related activity.
   (a) Choose one of the following topic areas:

      1. dodgeball (sports)
      2. business cycles (economics)
      3. carbon cycle (climate)
      4. models from data (data science)
      5. political polarization (politics)
      6. resource modeling (biology/ecology)

   (b)
      - Read the title and abstract.
      - Based on those, discuss with the group what you think the paper will be about.
      - Read (or skim) the introduction.
      - What have you learned about the big picture of where this paper fits in?
      - Look through the figures.
      - What did you learn from them?
      - Try to find the dynamical systems analysis in the paper.
        What are they doing? *bifurcations? statistics? time series?*
      - Develop an advertisement for what would make it interesting to work on this paper.

      We will change the groups to present the advertisements.

2. (Working in Cartesian with a system in polar)

   Using polar coordinates can be a straightforward way to make a system that is constructed to have isolated closed trajectories.

   Consider the system

$$\dot{r} = r(1-r)(2-r)$$
$$\dot{\theta} = 1.$$

   Although this looks like a system with one angular variable and one variable on the real line, instead we will interpret it as a polar description of a system defined in the xy-plane.

   (a) We think about this system in the xy-plane. Note that $\dot{\theta} = 1$ so $\theta(t) = (t + \theta_0)$, and we think of it mod 2π.

   The angle is continuously increasing in the counterclockwise direction. This continuous motion in the angle means that, away from the origin, there can't be a fixed point.

      - What kind of trajectory will you have for this system when $\dot{r} = 0$?
      - What about if $\dot{r} = 0$ and $r = 0$?

   (b) To analyze the system, first consider the 1d system

$$\frac{dx}{dt} = x(1-x)(2-x).$$

   What are the fixed points? Sketch what is happening on the x-axis. Restrict yourself to $x \geq 0$.

(c) Now think about the system

$$\dot{r} = r(1-r)(2-r)$$
$$\dot{\theta} = 1.$$

Show that $r = 1$ and $r = 2$ are phase curves of the system. Trajectories that start on these curves stay on these curves for all time.

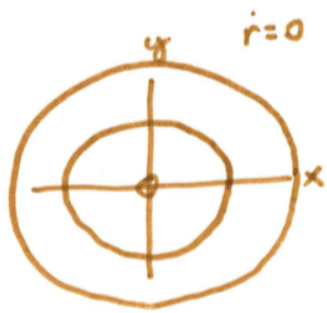
To do this, show that $\frac{dr}{dt} = 0$ when $r = 1$ or when $r = 2$. Why is this sufficient to show that these curves are phase curves?

(d) In the xy-plane, sketch in the $\dot{r} = 0$ trajectories. Try to sketch a phase portrait.

Answers:

1. a: $\dot{r} = 0$ and $\dot{r} = 0$ is a fixed point; just $\dot{r} = 0$ is a closed orbit.

b: fixed points at $0, 1, 2$. At $x = 3$, $\frac{dx}{dt} = 3(-2)(-1) > 0$ so 2 is unstable, 1 is stable, 0 is unstable.



c: $\dot{r} = r(1-r)(2-r)$. On $r = 1$ and on $r = 2$, $\dot{r} = 0$. So $r = 1$ and $r = 2$ are invariant curves (phase curves).

d: see above.

3. (A little more polar)

Consider the system

$$\dot{r} = r(2 - \sin\theta - r)$$
$$\dot{\theta} = 1.$$

Show that the curve $r = 2 - \sin\theta$ is not a phase curve of the system.

Trajectories that start on this curve will leave it.

4. Consider the system

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= -x - y(x^2 + y^2 - 1).\end{aligned}$$

(a) Transform the system to polar coordinates.

(b) Show that there is an invariant set at $r = \sqrt{x^2 + y^2} = 1$.